Muhammad Talha Khan

Date of birth: 9 Jul 2003 | **Nationality:** Pakistani | **Gender:** Male | **Phone number:** (+92) 03149133858 (Mobile) | **Email address:** mtalhak9753@gmail.com | **LinkedIn:** Muhammad Talha Khan |

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WORK EXPERIENCE

ASSISTANT RESEARCHER – SECURED IOT DEVICES (SID) LAB, UET PESHAWAR – 4 Dec 2024 – Current – PESHAWAR, PAKISTAN

- Leading the AIoT-Based Smart Climate-Aware Greenhouse System for Sustainable Agriculture (Final Year Project, 1st Position at FYP Exhibition 2026): designed a fully autonomous, AI-driven greenhouse combining ESP32 sensor nodes with on-device Raspberry Pi 4 edge AI for predictive forecasting, NDVI-based plant health monitoring (>90% precision), and closed-loop control of irrigation, ventilation, and lighting.
- Delivering 30–50% lower water use, 20–40% higher crop yield, and 15–20 labor-hours saved per farm per week; contributing to wider IoT security research and serving as a STEM instructor at a KP Higher Education Department STEAM workshop (Apr 2026).

PROJECTS

12 APR 2024 – 4 MAY 2026

AIoT-Based Smart Climate-Aware Greenhouse System for Sustainable Agriculture (Final Year Project — 1st Position, FYP Exhibition 2026)

Designed and led a fully autonomous, AI-driven greenhouse system integrating an IoT sensor network with edge AI to address inconsistent climate control, water wastage, and late detection of plant stress. ESP32 microcontrollers streamed temperature, humidity, soil moisture, CO_2, light intensity, and live camera feed data, while a Raspberry Pi 4 performed on-device predictive forecasting, anomaly detection, and closed-loop control of irrigation, ventilation, grow lights, and shading. Implemented NDVI-based computer vision for plant health monitoring at over 90% precision. Results included 30–50% lower water use, 25–30% lower energy use, 20–40% higher crop yield, and 15–20 labor-hours saved per farm per week.

Link https://github.com/mtalhak003/Ai-Agri-Smart-GreenHouse-System

15 AUG 2025 – 5 SEP 2025

Online Learning Platform

Built a full-stack learning platform using PHP, Laravel, and MySQL with role-based authentication for students, instructors, and admins. Implemented course enrollment and progress tracking functionality.

Link https://github.com/mtalhak003/olp

10 APR 2025 – 1 MAY 2025

Smart Delivery Box

Developed a 12V battery-powered, temperature-controlled delivery box with heating and fan-based ventilation. Integrated IoT-based monitoring for real-time temperature and airflow control to extend safe delivery time for temperature-sensitive goods.

5 SEP 2024 – 10 FEB 2025

3D Quadcopter Drone

Designed the complete 3D body of a quadcopter, optimizing weight distribution, battery placement, and propeller sizing. Calculated flight-weight and thrust requirements to ensure stable flight performance.

28 JUL 2024 – 10 AUG 2024

Cross-OS Bridge

Developed a cross-platform application (Windows/Linux/Ubuntu) enabling real-time chat, secure file transfer, and AI-chatbot integration, using network and socket programming.

Link https://github.com/mtalhak003/CROSS-OS-BRIDGE

7 JAN 2024 – 10 FEB 2024

Interactive Racing Monkey with ESP-32 Control

Development of an interactive racing monkey robot which climbs on rope controlled by ESP32 and a mobile app via Bluetooth. Programmed ESP32 for mobile app integration, enabling intuitive robot control and immersive user experience. Detailed project documentation is available on my GitHub repository.

EDUCATION AND TRAINING

10 OCT 2022 – 7 JUL 2026 Peshawar, Pakistan
BACHELOR OF SCIENCE University of Engineering and Technology

Systems-2

Website https://www.uetpeshawar.edu.pk/cse.php | **Field of study** Computer Systems Engineering

3 SEP 2020 – 26 JUN 2022 Peshawar, Pakistan
HIGHER SECONDARY SCHOOL Edwards College Peshawar

• Physics • Mathematics • Chemistry • English Literature

Website https://edwardes.edu.pk/Home | **Field of study** Pre-Engineering | **Final grade** A

10 MAR 2007 – 20 APR 2020 Peshawar, Pakistan
SECONDRY SCHOOL Peshawar Model School Boys 2

Website https://peshawarmodel.edu.pk/pms/summer/b2.html

SKILLS

Programming Languages

C/C++ | MEAN/MERN | Javascript | Objective-C | Python (computer programming) | Java (computer programming) | MySQL | C# | computer vision | Intermediate knowledge of C C++ MATLAB FPGA(VHDL) | Experience with PHP, HTML, Bootstrap, Javascript and Laravel framework | Artificial Intelligence and Automation in robotics

Development Tools & Environments

Microsoft Office | Visual C - Visual Studio | GitHUB | GIT | Proteus | AutoCAD | Arduino IDE | Adobe Photoshop | MATLAB | Canva

Embedded Systems & Microcontrollers

Raspberry Pi | FPGA(VHDL - basic experience with Xilinx ISE) | Microcontrollers | ESP32 | Arduino | ESP32-S | communication Protocols(UART,SPI,I2C,CAN) | sensors

LANGUAGE SKILLS

Mother tongue(s): **ENGLISH** | **PASHTO** | **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
GERMAN	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

REFERENCES

Available on Request

MANAGEMENT AND LEADERSHIP SKILLS (VOLUNTEER)

16 OCT 2022 – 7 JUL 2026
Director of Operations, Computer Society, UET Peshawar

Managed operations for 50+ major events including university-wide activities and international conferences, streamlining workflows and strengthening team performance across multiple venues.

Link https://www.linkedin.com/company/computersocietyuetp/

SEP 2025 – 7 JUL 2026
Embedded Systems Lead, Google Developer Groups on Campus (GDGoC), UET Peshawar

Led embedded systems initiatives, including hands-on workshops, hardware sessions, and IoT training, contributing to a 35% increase in student participation across embedded, AI, and software domains.

2025 – JUL 2026

Director of Judges Support, Hult Prize, UET Peshawar

- Coordinated judge panels, managed schedules, and facilitated communication between participants and evaluation teams.